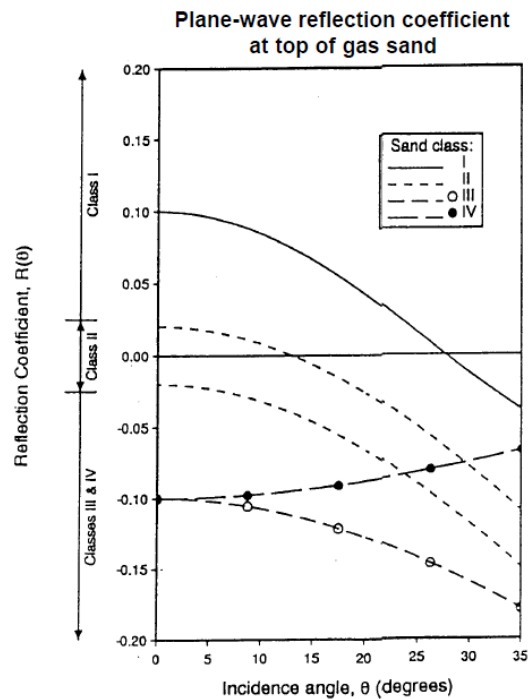


Indian Institute of Technology (Indian School of Mines), Dhanbad
End Semester Examination (Monsoon Semester), Academic Year 2023-24
Seismic Data Processing and Interpretation (GPC 517)

Model Answer

Q.1

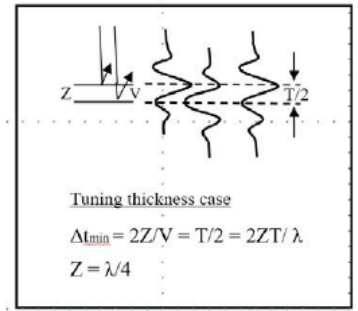
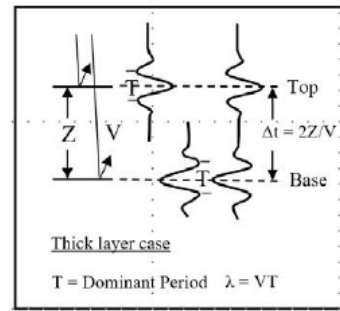
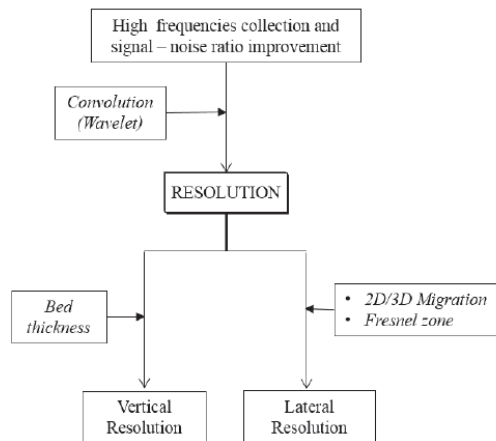
Class	Relative Impedance	Quadrant	A	B	Amplitude vs. Offset
I	Higher than overlying unit	IV	+	-	Decreases
II	About the same as the overlying unit	II, III, or IV	+ or -	-	Increase or decrease; may change sign
III	Lower than overlying unit	III	-	-	Increases
IV	Lower than overlying unit	II	-	+	Decreases



Class IV gas sand on a figure shows their gas-sand classification based on normal incidence reflection coefficient. The vertical axis is reflection coefficient and the horizontal axis is local angle of incidence. Note that Class III and IV gas sands may have identical normal incidence reflection coefficients, but the magnitude of Class IV sand reflection coefficients decreases with increasing angle of incidence while Class III reflection coefficient magnitudes increase.

Marks distribution: Correct demonstration with all points and line diagram – 6 & 4; Partly explanation with untidy/wrong line diagram – 1 to 4 & 1 to 3; wrong explanation with wrong line diagram – 0.

Q.2.



Three types of seismic distortions are of most concern:

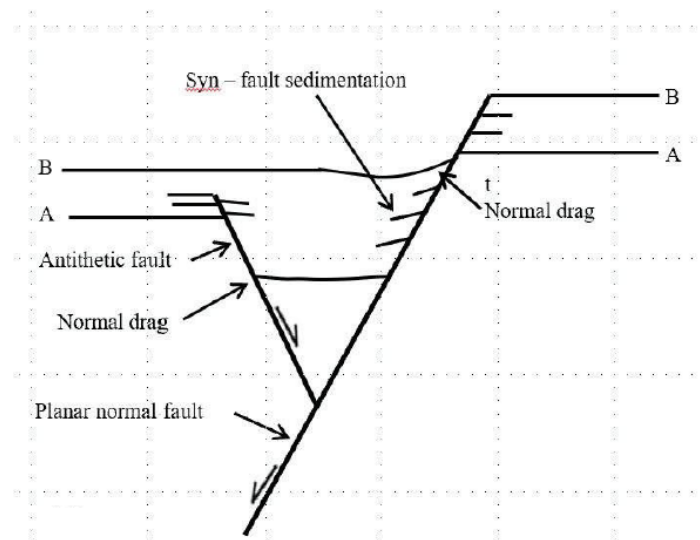
1. Events are not shown in their true positions, in time or depth,
2. Reflection response differences cause apparent bedding changes.
3. Noise (extraneous events) interferes with and masks signal.

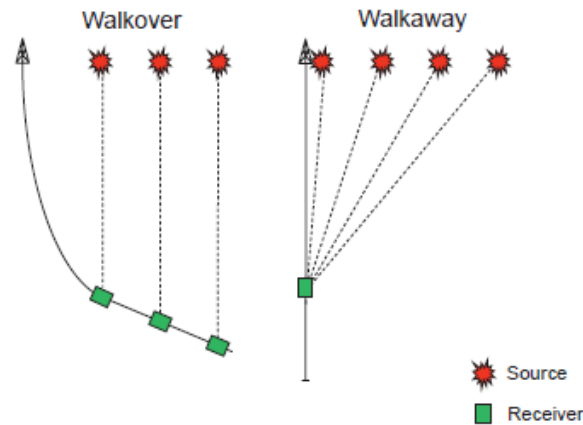
Mechanisms that produce distortion include:

- Focusing
- Shadow zones
- Subsurface discontinuities
- Inadequate resolution
- Topography and near-surface variations
- Velocity changes
- Noise generators

Marks distribution: Correct demonstration with all points and line diagram – 5 & 5; Partly explanation with untidy/wrong line diagram – 1 to 3 & 1 to 3; wrong explanation with wrong line diagram – 0.

Q.3.





Marks distribution: Correct demonstration with all points and line diagram – 7 & 3; Partly explanation with untidy/wrong line diagram – 1 to 4 & 1 to 2; wrong explanation with wrong line diagram – 0.

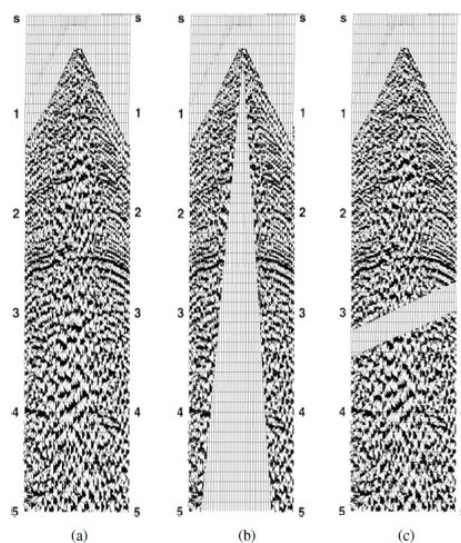
Q.4.

Front-End Mute:

The far geophone groups are quite distant from the energy source. On the traces from these receivers, refractions may cross and interfere with reflection information from shallow reflectors. However, the nearer traces are not so affected. When the data are stacked, the far traces are muted (zeroed) down to a time at which reflections are free of refractions. The mute schedule is a set of time, trace pairs that define the end of the muting. Mute changes the relative contribution of the components of the stack as a function of record time. In the early part of the record, the long offset may be muted from the stack because the first arrivals are disturbed by refraction arrivals, or because of the change in their frequency content after applying normal moveout. The transition where the long offsets begin to contribute may be either gradual or abrupt. However, an abrupt change may introduce frequencies that will distort the design of the deconvolution operator.

Surgical Mute:

Muting may be over a certain time interval to keep ground roll, airwave, or noise patterns out of the stack. This is especially applicable if the noise patterns are in the same frequency range as the desired signal. Convolution to filter out the noise may also attenuate the desired signal.



Surgical mutes. (a) record without surgical mute. (b) and (c) with surgical mutes applied

Front-end mute is applied at the start of a record while a surgical mute is applied somewhere in the body of a record. The front-end mute is applied to eliminate the first breaks and high amplitude events following them. These events provide no subsurface information. Surgical mutes are applied to eliminate noise trains that interfere with signal.

Static Corrections:

Two main types of correction need to be applied to reflection times on individual seismic traces in order that the resultant seismic section gives a true representation of geological structure. These are dynamic and static corrections. The static correction is so-called because it is a fixed time correction applied to the entire trace. A dynamic correction varies as a function of time as in application of NMO corrections.

In order to obtain a seismic section that accurately shows the subsurface structure, the datum plane at a known elevation above mean sea level and below the base of the variable-velocity weathered layer. The value of the total statics (ΔT) depends on the following factors:

1. The perpendicular distance from the source to the datum plane.
2. Surface topography; that is, the perpendicular distance from the geophone to the datum plane.
3. Velocity variations in the surface layer along the seismic line.
4. Irregularities in thickness of the near-surface layer.

In computing ΔT it is usually assumed that the reflection ray path in the vicinity of the surface is vertical.

The total correction is:

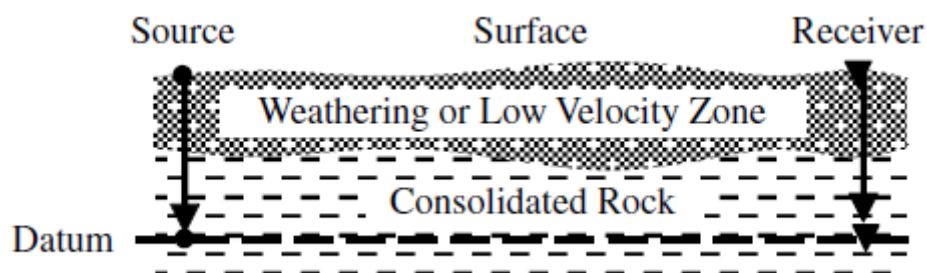
$$\Delta T = \Delta t_s + \Delta t_r$$

where

Δt_s = the source correction, in ms

Δt_r = the receiver correction, in ms

As indicated in Fig. elevation or datum statics effectively move source and receiver from the surface to the datum plane.



Static corrections

Q.5. Starting from,

$$R(\theta) = A + B \sin^2 \theta + C \sin^2 \theta \tan^2 \theta,$$

where

$$A = \frac{1}{2} \left(\frac{\Delta V_p}{V_p} + \frac{\Delta \rho}{\rho} \right)$$

$$B = \frac{\Delta V_p}{2 V_p} - 4 \left(\frac{V_s}{V_p} \right)^2 \left(\frac{\Delta V_s}{V_s} \right) - 2 \left(\frac{V_s}{V_p} \right)^2 \left(\frac{\Delta \rho}{\rho} \right)$$

$$\text{and } C = \frac{1}{2} \frac{\Delta V_p}{V_p},$$

$$\text{where } V_p = \frac{V_{p1} + V_{p2}}{2}, \quad V_s = \frac{V_{s1} + V_{s2}}{2}, \quad \rho = \frac{\rho_1 + \rho_2}{2},$$

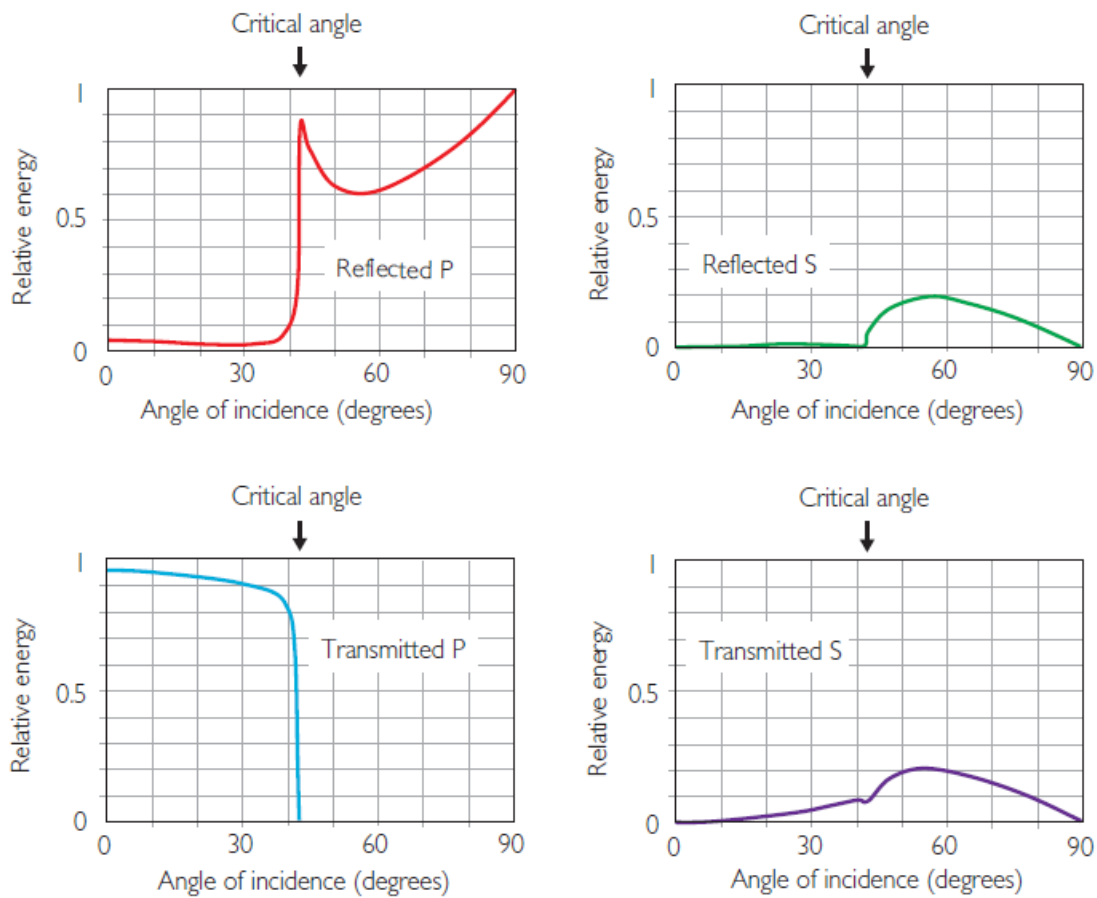
$$\left(V_s / V_p \right)^2 = \frac{(V_{s1} / V_{p1})^2 + (V_{s2} / V_{p2})^2}{2},$$

$$\Delta V_p = V_{p2} - V_{p1}, \quad \Delta V_s = V_{s2} - V_{s1},$$

$$\text{and } \Delta \rho = \rho_2 - \rho_1.$$

The first term (A) is the zero angle reflection coefficient related to the contrast of acoustic impedance, whilst the second term (B) introduces the effect of shear velocity at non-zero angles. A third term (C) determines the curvature of the amplitude response near to the critical angle.

Final outcome.

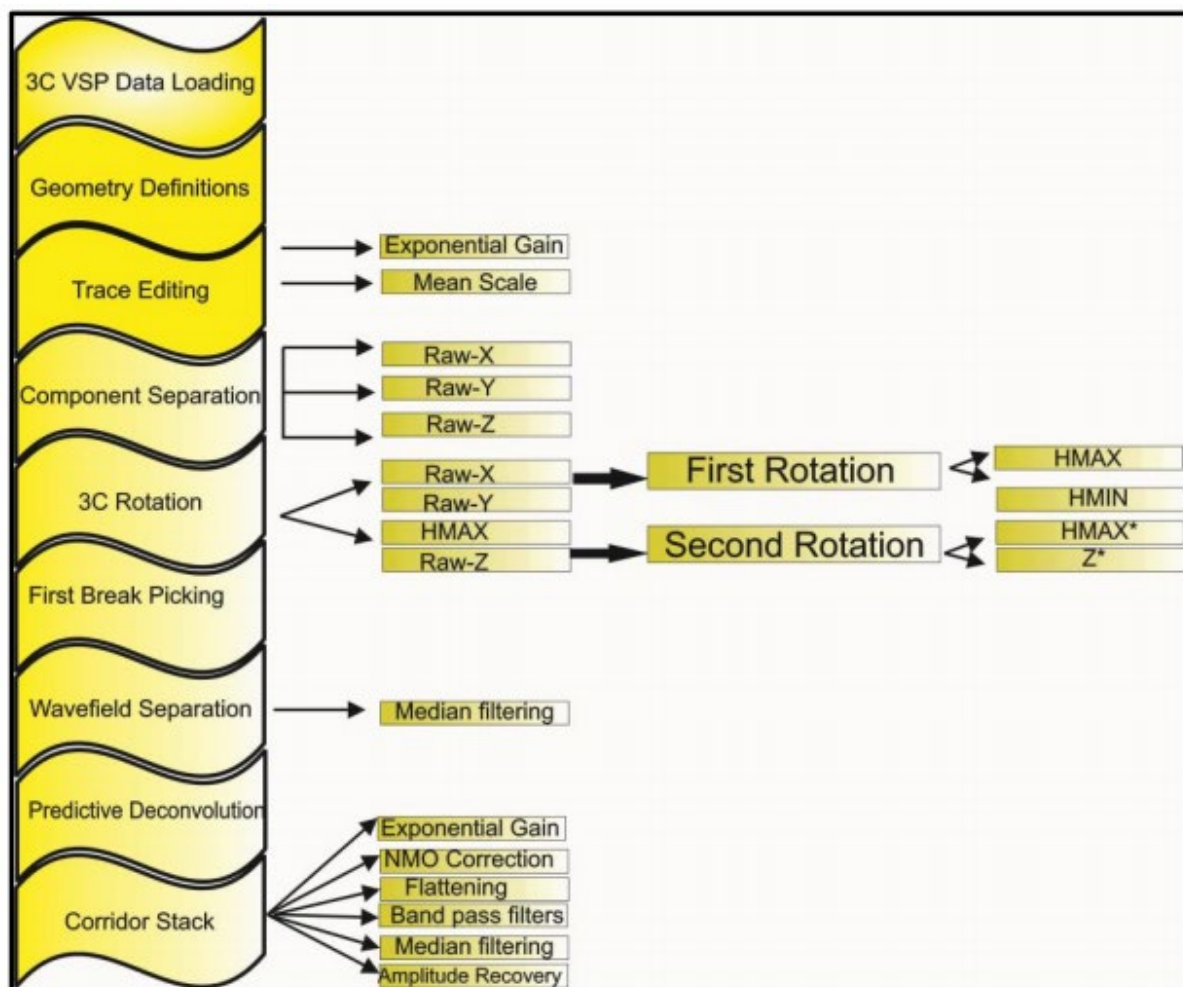


Estimated values are required to be mentioned in the calculation.

Marks distribution: Correct demonstration with all points and line diagram – 10; Partly explanation with untidy/wrong line diagram – 1 to 5; wrong explanation with wrong line diagram – 0.

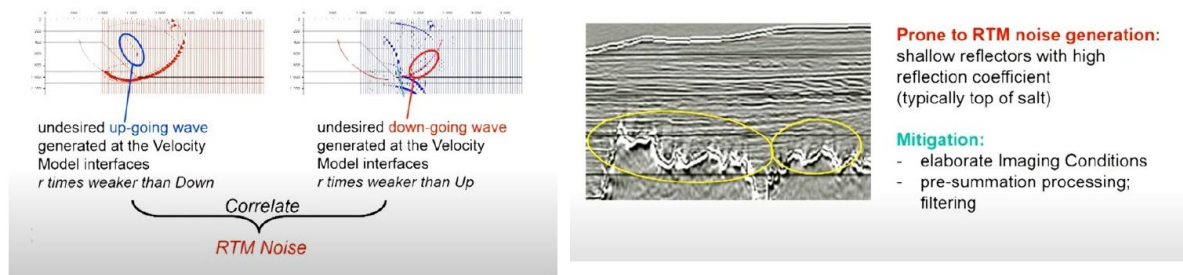
Q.6.

1. Reflections can be tied directly from the seismic record to the well log.
2. Multiples can be easily identified
3. Faults can be detected by offsets of events on the VSP.
4. Reflections beneath the TD can be evaluated.
5. Reflection coefficients can be calculated accurately.
6. Detailed interval velocity can be calculated.
7. An evaluation of what the surface seismic section can and cannot resolve can be made.



Marks distribution: Correct demonstration with all points and line diagram – 5 & 5; Partly explanation with untidy/wrong line diagram – 1 to 3 & 1 to 3; wrong explanation with wrong line diagram – 0.

Q.7.



Grid sampling must be small enough to avoid aliasing and numerical instability

wavelength $\lambda_{\min} = \frac{\text{lowest local Velocity}}{\text{maximum frequency}}$

sampling $\leq \lambda_{\min} / 10$

↓

Computing effort (3D) $\sim (\text{max. freq.})^4$

10 x 10 km aperture and 8 km depth case

Migration time = 6s ; $V_{\min} = 1500\text{m/s}$

Maximum frequency = 20 Hz

Maximum frequency = 80 Hz

$\lambda_{\min} = 75\text{m}$; dx, dy, dz $\leq 7.5\text{m}$; dt=2ms

$\lambda_{\min} = 19\text{m}$; dx, dy, dz $\leq 1.9\text{m}$; dt=1ms

Storage for 1 Shot:
3000 snapshots = 23 Tbytes
Computation: N operations

Storage for 1 Shot:
6000 snapshots = 2800 Tbytes
Computation: $256 \times N$

Schematic diagrams for RTM Noise related to up-going, down-going waves and Migration related noise are required.

The RTM workflow

⇒ For one Shot :

1. Compute the down-going wavefield
2. Reconstruct the up-going wavefield
3. "Compare" them at each snapshot time to get the contribution of that time to the final image
4. Accumulate the results over time to get one migrated shot

Marks distribution: Correct demonstration with all points and line diagram – 6 & 4; Partly explanation with untidy/wrong line diagram – 1 to 3 & 1 to 2; wrong explanation with wrong line diagram – 0.

Q.8.

First, let us examine the required time sampling, Δt , to avoid temporal aliasing. We want to ensure that the maximum frequency, 100 Hz (F_N), is sampled properly. Applying equation 1, we find

$$\Delta t \leq \frac{1}{2F_N} = \frac{1}{2(100)} = 5 \text{ ms}.$$

Therefore, a time-sample interval of 5 ms or smaller is required. Most conventional surveys are recorded at 4 ms or less, so there is little concern for temporal aliasing.

Let us examine the required geophone spacing, Δx , to avoid spatial aliasing. We must consider the limiting case in which $V = 2500 \text{ m/s}$, $f = 100 \text{ Hz}$, and $\theta = 40^\circ$. Applying equation 2, we find

$$\Delta x \leq \frac{V}{4f \sin \theta} = \frac{2500}{4(100) \sin 40^\circ} = 9.7 \text{ m}.$$

Therefore, a maximum geophone spacing of 9.7 m (essentially 10 m) is required to image all expected dips for the given medium velocity and bandwidth. If the spacing is not feasible, processing techniques such as filtering or trace interpolation, as discussed above, may be necessary.

Generally, surface recordings of seismic data are in the 5–100-Hz range, and sonic logs are in the 10–12-kHz range for monopoles sources. If the velocity were not dispersive (not frequency dependent), we might expect nearly perfect correlations. In some cases, as in the examples, the correlations are very good and worthwhile.

However, in other cases, this match is not perfect. There are several reasons for this, including:

- 1) velocity dispersion
- 2) problems with sonic logs, such as cycle skipping or mud invasion
- 3) noisy seismograms
- 4) nonzero-phase wavelets
- 5) polarity differences between synthetic seismograms and seismic data

Marks distribution: Correct demonstration with all points and line diagram – 6 & 4; Partly explanation with untidy/wrong line diagram – 1 to 3 & 1 to 2; wrong explanation with wrong line diagram – 0.

Q.9.

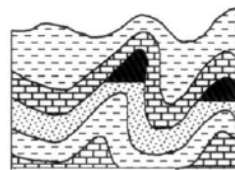


Structural Interpretation

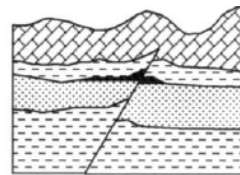
- Reflector correlation
- Fault terminations
- Tying seismic lines together and to well control
- Picking of reflector times
- Contour mapping
- Depth conversion using velocity controls

Marks distribution: Correct demonstration with all points and line diagram – 6 & 4; Partly explanation with untidy/wrong line diagram – 1 to 3 & 1 to 2; wrong explanation with wrong line diagram – 0.

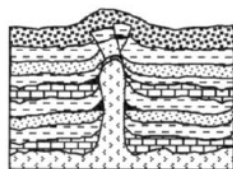
Q.10.



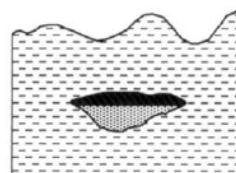
a. Anticline



b. Normal Fault



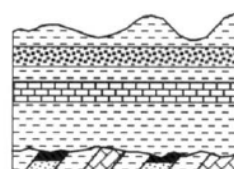
c. Salt Dome



d. Sand Lens



e. Reef



f. Angular Unconformity

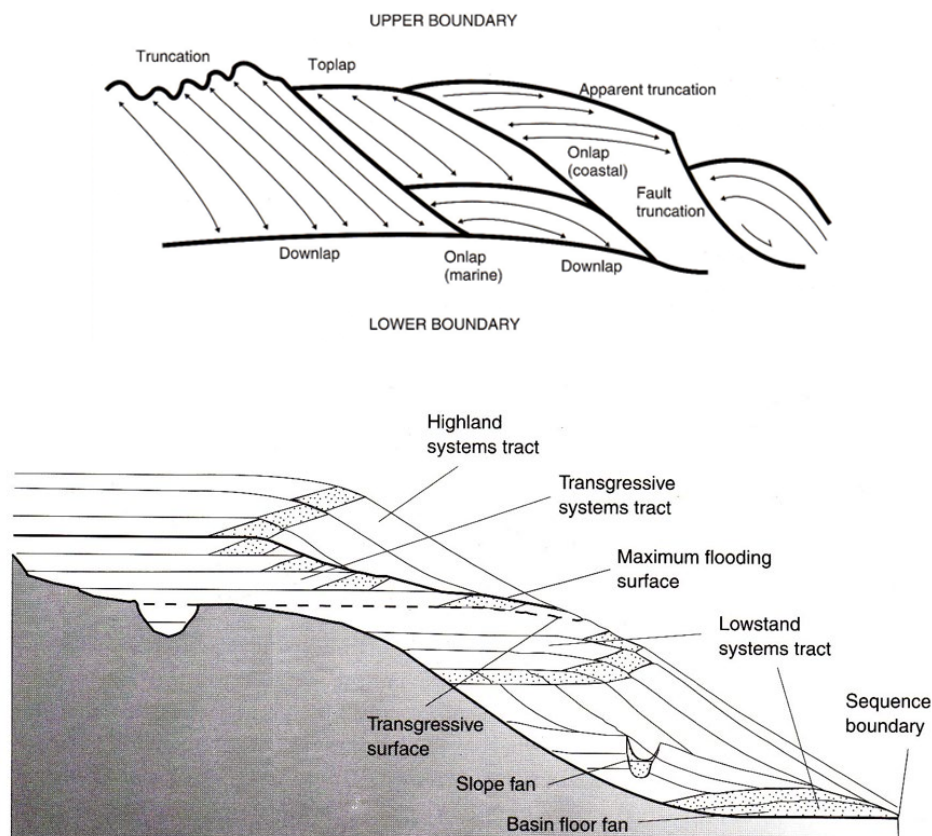
Long period multiples can be suppressed by parabolic Radon transform based on the moveout differences between primary and multiple reflection hyperbolas in CDP gathers. It is accomplished by an application of the τ -p transform to NMO-corrected CDPs. The most important advantage of the method is that it does not require information on the formation mechanism of the multiples. Practical applications indicate that the Radon velocity filters are effective in suppressing the long period multiples, especially at far offsets. However, for complete success, the residual moveout between primary and multiple reflections after NMO correction should be at least

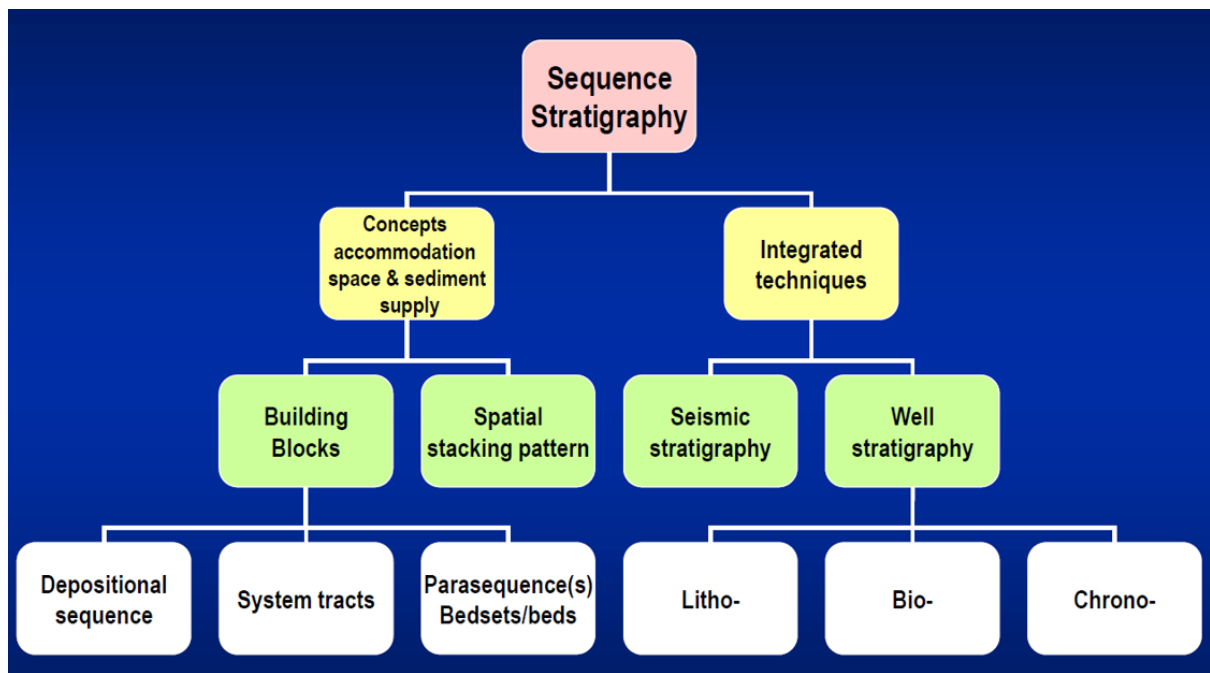
30 ms. In practice, the higher the residual moveout, the more effective the method in multiple elimination. This prerequisite requires long offset seismic data to be collected, and therefore the method is generally almost ineffective on the data collected with a short streamer cable. In addition, since the method is based on the moveout differences after NMO correction, it also requires a careful velocity analysis.

The amplitudes in the CDP gathers are mapped into different areas in the τ -p domain, depending on their curvature controlled by their residual moveout times since the horizontal axis of the τ -p gather corresponds to residual moveouts. Providing an accurate velocity field, the multiple reflections are still hyperbolic while the primary reflections are flattened following the NMO correction. This difference results in the amplitudes of primaries and the multiples being mapped in different areas in the τ -p domain, in which the amplitudes of the multiples are muted out (Landa et al., 1999). The important point here is the correct picking of this mute zone, which discriminates the primary and multiple amplitudes in the τ -p domain. The method involves the following steps.

- i. NMO correction with velocities of primary reflections is applied to input CDP.
- ii. CDP gather is transferred into the τ -p domain by Radon transform.
- iii. The area comprising the multiple amplitudes in τ -p domain is muted out.
- iv. CDP is transferred back into the time distance domain by inverse Radon transform.
- v. If required, an inverse NMO is applied to obtain original CDP gather.

Or





Marks distribution: Correct demonstration with all points and line diagram – 6 & 4; Partly explanation with untidy/wrong line diagram – 1 to 3 & 1 to 2; wrong explanation with wrong line diagram – 0.